

CHAPTER II.4, WEEK 5

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Assume all schemes are noetherian, as otherwise everything breaks.

4.1. Show that a finite morphism is proper.

Proof. Let $f : X \rightarrow Y$ be a finite morphism. Take an open affine cover $\{V_i = \text{Spec } B_i\}$ of Y with each $f^{-1}(B_i) = \text{Spec } A_i$ a finitely generated B_i -module. It suffices to consider the case $f|_{\text{Spec } A_i} : \text{Spec } A_i \rightarrow \text{Spec } B_i$, so we may assume $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a morphism of affine schemes corresponding to a ring morphism $f : A \rightarrow B$.

$f^* : \text{Spec } B \rightarrow \text{Spec } A$ is finite, so in particular finite type. Now fix a valuation ring R with $R \hookrightarrow K$ its fraction field, and morphisms $u : B \rightarrow K$, $v : A \rightarrow R$. We are in the affine case, so (4.7) gives us that f^* is proper iff there is a unique morphism $h : B \rightarrow R$ such that the following diagram commutes.

$$\begin{array}{ccc} K & \xleftarrow{u} & B \\ \uparrow i & \nearrow h & \uparrow f \\ R & \xleftarrow{v} & A \end{array}$$

B is a finite A -module, so in particular it is integral over A . Consider the inclusion $v(A) \hookrightarrow u(B)$, and let $u(b) \in u(B)$. Take $a_1, \dots, a_n$ such that

$$\begin{aligned} b^n + f(a_1)b^{n-1} + \dots + f(a_n) &= 0 \\ \implies u(b)^n + uf(a_1)u(b)^{n-1} + \dots + uf(a_n) &= 0 \\ \implies u(b)^n + v(a_1)u(b)^{n-1} + \dots + v(a_n) &= 0, \end{aligned}$$

whence our $u(b)$, and thus all of $u(B)$, is integral over $v(A)$. Valuation rings are integrally closed, so it follows that $u(B) = v(A) \subset R$. Then $h = u : B \rightarrow R$ is the unique map satisfying the diagram, whence f^* is proper. $\square$

4.2. Let S be a scheme. Let f and g be two S -morphisms from a reduced S -scheme X to a separated S -scheme Y which agree on an open dense subset of X . Show that $f = g$. Give examples to show that this result fails if either (a) X is nonreduced, or (b) Y is nonseparated. [*Hint:* Consider the map $h : X \rightarrow Y \times_S Y$ obtained from f and g .]

Proof. Let $U \subset X$ be an open dense subset on which f and g agree, $i : U \hookrightarrow X$ the inclusion. Denote $\Delta : Y \rightarrow Y \times_S Y$, and $h : X \rightarrow Y \times_S Y$ the map given by $f, g : X \rightarrow Y$ and the universal property of $Y \times_S Y$. View X and Y as $Y \times_S Y$ -schemes using the aforementioned maps and take the fibred product $Z := X \times_{Y \times_S Y} Y$.

$$\begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow v & & \downarrow h \\ Y & \xrightarrow{\Delta} & Y \times_S Y \end{array}$$

(U, f, i) completes the pullback square as well, so we have the following diagram.

$$\begin{array}{ccc}
 U & \longrightarrow & U \\
 \downarrow & & \downarrow i \\
 Z & \longrightarrow & X \\
 \downarrow v & & \downarrow h \\
 Y & \xrightarrow{\Delta} & Y \times_S Y
 \end{array}$$

Y separated means Δ is a closed immersion, and closed immersions are preserved under base extension by 3.11a, so $Z \rightarrow X$ is a closed immersion. Without loss of generality, we assume Z is in fact a closed subscheme of X .

U inherits its structure sheaf from X , and is thus reduced, whence the scheme theoretic image of $i : U \rightarrow X$ is simply X . The diagram gives us that $i : U \rightarrow X$ factors as $U \rightarrow Z \rightarrow X$, so the identity $\text{id} : X \rightarrow X$ factors through $Z \rightarrow X$. That is, we have a composition $\text{id} : X \rightarrow Z \rightarrow X$, whence $Z = X$. Then $h = \Delta \circ v$, and the following diagram tells us that $v = f$.

$$\begin{array}{ccccc}
 & & f & & \\
 & & \curvearrowright & & \\
 X & \xrightarrow{v} & Y & \xrightarrow{\Delta} & Y \times_S Y & \xrightarrow{\text{id}} & Y \\
 & \searrow h & & & & \nearrow & \\
 & & & & & &
 \end{array}$$

We conclude that $f = g$ as morphisms on X . □

4.5. Let X be an integral scheme of finite type over a field k , having function field K . We say that a valuation of K/k has *center* x on X if its valuation ring R dominates the local ring $\mathcal{O}_{x,X}$.

- (a) If X is separated over k , then the center of any valuation of K/k on X (if it exists) is unique.
- (b) If X is proper over k , then every valuation of K/k has a unique center on X .
- (c) (*) Prove the converses of (a) and (b).
- (d) If X is proper over k , and if k is algebraically closed, show that $\Gamma(X, \mathcal{O}_X) = k$.
[Hint: Let $a \in \Gamma(X, \mathcal{O}_X)$ with $a \notin k$. Show that there is a valuation ring R of K/k with $a^{-1} \in \mathfrak{m}_R$. Then use (b) to get a contradiction.]

Proof. (a) Take a valuation of K/k with valuation ring R . Let $\xi \in X$ be the generic point of X , so that $\{\xi\}^- = X$ with the reduced induced structure. We have $R \supset k$ by definition of a valuation of K/k . With notation as in (4.3) and (4.7), fixing the map $U \rightarrow X$ with image ξ gives us the following commutative square.

$$\begin{array}{ccc}
 U & \longrightarrow & X \\
 \downarrow & & \downarrow \\
 T & \longrightarrow & \text{Spec } k
 \end{array}$$

$k(\xi) = K$, so, by Lemma 4.4, diagonal maps $h : T \rightarrow X$ so that the diagram commutes are in correspondance with points $x \in \{\xi\}^- = X$ such that R dominates $\mathcal{O}_{x,X}$. This is exactly

the condition that x is a center of the valuation, so $X \rightarrow \operatorname{Spec} k$ separated implies centers are unique when they exist.

(b) By (a), centers of a valuation of K/k are in correspondance with diagonals of the above diagram. $X \rightarrow \operatorname{Spec} k$ implies there exists a unique such diagonal, whence a unique center.

(d) The map $X \rightarrow \operatorname{Spec} k$ gives us an inclusion of rings $k \subset \Gamma(X, \mathcal{O}_X) =: A$ by Exercise 2.4. Suppose that $A \supsetneq k$, and take $a \in A \setminus k$. Consider the ring $B := k[a^{-1}] \subset A$. k algebraically closed implies $a^{-1} \notin B^\times$, so we can find a prime ideal $\mathfrak{p} \in \operatorname{Spec} B$. Then we have a local ring

$$B_{\mathfrak{p}} \subset B_{(0)} \subset A_{(0)} = K$$

with maximal ideal $\mathfrak{m}_{\mathfrak{p}} := \mathfrak{p}B_{\mathfrak{p}}$. Take some local ring $R \subset K$ which dominates $B_{\mathfrak{p}}$ and is maximal with respect to domination, so that R is a valuation ring with $a^{-1} \in \mathfrak{m}_{\mathfrak{p}} \subset \mathfrak{m}_R$. Additionally, $k \subset B \subset R$, so R is indeed a valuation of K/k .

Now, suppose that $X \rightarrow \operatorname{Spec} k$ is proper. We receive a unique $x \in X$ such that R dominates $\mathcal{O}_{x,X}$. Then $a \in \mathcal{O}_X \subset \mathcal{O}_{x,X} \subset R$ implies $a \notin \mathfrak{m}_R$, contradicting our construction of R . Hence, no such center exists, i.e. X is not proper over k by (b). Note that X may still be separated over k , as we have only contradicted existence, rather than uniqueness. $\square$

4.8. Let $\mathcal{P}$ be a property of morphisms of schemes such that:

- (i) a closed immersion has $\mathcal{P}$;
- (ii) a composition of two morphisms having $\mathcal{P}$ has $\mathcal{P}$;
- (iii) $\mathcal{P}$ is stable under base extension.

Then show that:

- (a) a product of morphisms having $\mathcal{P}$ has $\mathcal{P}$;
- (b) if $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two morphisms, and if $g \circ f$ has $\mathcal{P}$ and g is separated, then f has $\mathcal{P}$;
- (c) If $f : X \rightarrow Y$ has $\mathcal{P}$, then $f_{\text{red}} : X_{\text{red}} \rightarrow Y_{\text{red}}$ has $\mathcal{P}$.

[Hint: For (b), consider the graph morphism $\Gamma_f : X \rightarrow X \times_Z Y$, and note that it is obtained from the diagonal morphism $\Delta : Y \rightarrow Y \times_Z Y$.]

Proof. (a) Let $f : X \rightarrow Y$ and $g : X' \rightarrow Y'$ be morphisms having $\mathcal{P}$. Transitivity of base extensions gives us that

$$\begin{aligned} X \times_Y (Y \times_S X') &\cong X \times_S X' \\ (Y \times_S Y') \times_{Y'} X' &\cong Y \times_S X', \end{aligned}$$

so we can construct the following diagram.

$$\begin{array}{ccccc} X' & \xrightarrow{\operatorname{id}_{X'}} & X' & \xrightarrow{g} & Y' \\ \uparrow & & \uparrow & & \uparrow \\ X \times_S X' & \xrightarrow{f'} & Y \times_S X' & \xrightarrow{g'} & Y \times_S Y' \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y & \xrightarrow{\operatorname{id}_Y} & Y \end{array}$$

Both of f', g' have property $\mathcal{P}$, since it is stable under base extension. Additionally, their composition $f' \circ g'$ has property $\mathcal{P}$ by closure under composition. However, we see that $f' \circ g'$ is exactly the product of f, g by the above diagram, which gives us our result.

(b) The following isomorphism holds on open affines, hence globally.

$$(X \times_S Y) \times_{S'} Z \cong X \times_S (Y \times_{S'} Z)$$

Then, in our particular case, we have

$$(X \times_Z Y) \times_{Y \times_Z Y} Y \cong X \times_Z (Y \times_{Y \times}).$$

This allows us to create the following diagram of pullback squares.

$$\begin{array}{ccccc} & & X & \xrightarrow{g \circ f} & Z \\ & & \uparrow & & \uparrow g \\ X & \xrightarrow{\Gamma_f} & X \times_Z Y & \xrightarrow{\text{pr}_2} & Y \\ f \downarrow & & \downarrow (f, \text{id}) & & \\ Y & \xrightarrow{\Delta} & Y \times_Z Y & & \end{array}$$

Seperability of g implies Δ is a closed immersion, and thus has $\mathcal{P}$. Then stability under base extension gives us that Γ_f and pr_2 have $\mathcal{P}$, pulling back along the squares. However, we have that $\text{pr}_2 \circ \Gamma_f = f$, so that f has $\mathcal{P}$ by stability under composition.

(c) We have that $f : X \rightarrow Y$ induces a map $g : X \rightarrow Y_{\text{red}}$ with $f : X \xrightarrow{g} Y_{\text{red}} \xrightarrow{h} Y$, where h is the natural map. Note that h is a closed immersion, as $A \rightarrow A_{\text{red}}$ is surjective for any ring A . Then h is seperated and f has $\mathcal{P}$, so g has $\mathcal{P}$ by (b). Finally, $f_{\text{red}} : X_{\text{red}} \rightarrow X \xrightarrow{g} Y_{\text{red}}$ has property $\mathcal{P}$ since $X_{\text{red}} \rightarrow X$ is a closed immersion, and $\mathcal{P}$ is stable under composition. $\square$